

API	Instruction code			Operand						Function							
0103	D	/	P	$S_1 \cdot S_2 \cdot D$						Dividing binary numbers							

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1	●	●			●	●	●	●	●		○	○	○	○		
S_2	●	●			●	●	●	●	●		○	○	○	○		
D		●			●	●	●	●			○					

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1		●	●			●	●				●	●	
S_2		●	●			●	●				●	●	
D		●	●			●	●				●	●	

Pulse instruction	16-bit instruction	32-bit instruction
AS	AS	AS

Symbol

En	/	En	/P
S1	D	S1	D
S2		S2	

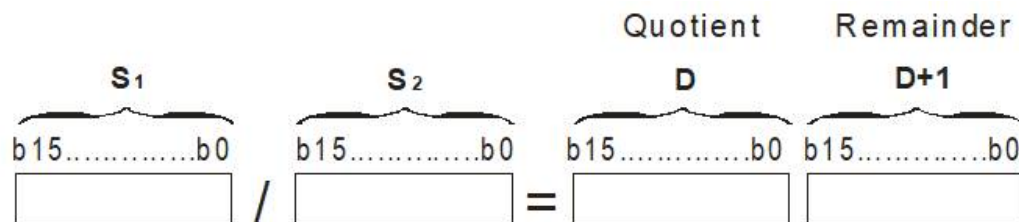
En	D/	En	D/P
S1	D	S1	D
S2		S2	

 S_1 : Dividend S_2 : Divisor

D : Quotient; remainder

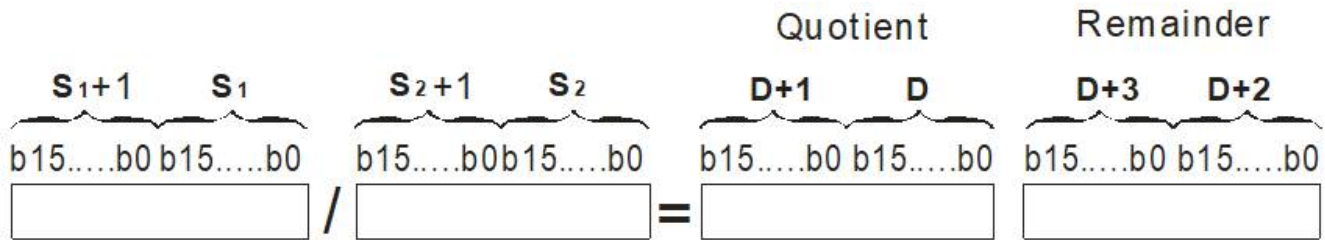
Explanation

- This instruction divides the signed binary value in S_1 by the signed binary value in S_2 , and stores the quotient and the remainder in D.
- Only the 32-bit instructions can use the 32-bit counter.
- When the sign bit is 0, the value is a positive one. When the sign bit is 1, the value is a negative one.
- Dividing 16-bit values:



The operand **D** occupies two consecutive devices. The quotient is stored in **D**, and the remainder is stored in **D+1**.

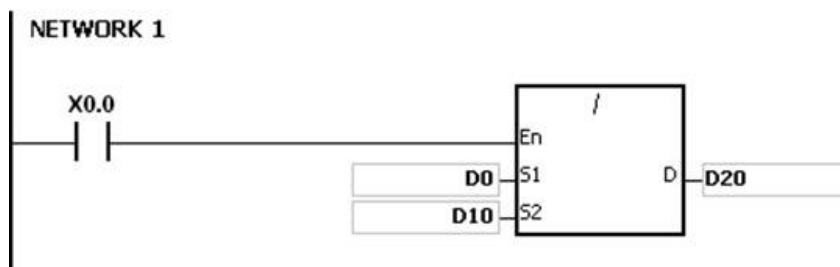
5. Dividing 32-bit values:



The operand **D** occupies two devices. The quotient is stored in (**D+1**, **D**), and the remainder is stored in (**D+3**, **D+2**).

Example

When X0.0 is ON, the instruction divides the dividend in D0 by the divisor in D10, and stores the quotient in D20, and stores the remainder in D21. Whether the result is a positive value or a negative value depends on the state of the highest bit.

**Additional remarks**

1. If the device exceeds the range, the instruction is not executed, SM0 is ON, and the error code in SR0 is 16#2003.
2. If the divisor is 0, the instruction is not executed, SM0 is ON, and the error code in SR0 is 16#2012.
3. If you declare the operand **D** used during the execution of the 16-bit instruction in ISPSOft, the data type is ARRAY [2] of WORD/INT.
4. If you declare the operand **D** used during the execution of the 32-bit instruction in ISPSOft, the data type is ARRAY [2] of DWORD/DINT.